$a = 1 \quad b = 2 \quad c = -2$

$$x = \frac{-2 \pm \sqrt{4 - 4 \times 1 \times -2}}{2 \times 1}$$

$$= \frac{-2 \pm \sqrt{12}}{2} = \frac{-2 \pm \sqrt{4 \times 3}}{2}$$

$$x = \frac{-2 \pm 2\sqrt{3}}{2}$$

$$= \frac{-2 + 2\sqrt{3}}{2}, \quad \frac{-2 - 2\sqrt{3}}{2}$$

$x = -1 + \sqrt{3}\ ; -1 - \sqrt{3}$

Since x is + ve,

$= -1 + 1.732$

$-1 - \sqrt{3}$ not possible

$= 0.732$

∴ Equilibrium concentration of

$$[PCl_5]_{eq} = \frac{1 - x}{1} = 1 - 0.732 = 0.268\ M$$

$$[PCl_3]_{eq} = \frac{x}{1} = \frac{0.732}{1} = 0.732$$

$$[Cl_2]_{eq} = \frac{x}{1} = \frac{0.732}{1} = 0.732$$

5. $N_2O_4(g) \rightleftharpoons 2NO_2(g)$

$T_1 = 298\ K \qquad K_{P_1} = 0.15$

$T_2 = 100^o\ C = 100 + 273 = 373\ K$ ;
$K_{P_2} = ?$

$$\log\left(\frac{K_2}{K_1}\right) = \frac{\Delta H^o}{2.303\ R}\left[\frac{T_2 - T_1}{T_1 T_2}\right]$$

$K_2 > K_1$ and $T_2 > T_1$

$$\log\left(\frac{K_{P_2}}{0.15}\right) = \frac{57.2\ KJ\ mol^{-1}}{2.303 \times 8.314\ JK^{-1}\ mol^{-1}}\left[\frac{373 - 298}{373 \times 298}\right]$$

$$\log\left(\frac{K_{P_2}}{0.15}\right) = \frac{57.2 \times 10^{+3} \times 75}{2.303 \times 8.314 \times 373 \times 298}$$

$$\log\left(\frac{K_{P_2}}{0.15}\right) = 2.02$$

$$\frac{K_{P_2}}{0.15} = 104.7$$

$$K_{P_2} = 104.7 \times 0.15$$

$$K_{P_2} = 15.705$$

## KEY FOR MCQs

1. $K_b = 0.8 \times 10^{-5}$

$K_f = 1.6 \times 10^{-4}$

$$K_{eq} = \frac{K_f}{K_b} = \frac{1.6 \times 10^{-4}}{0.8 \times 10^{-5}} = 20 \text{ (option (a))}$$

2. $$K_1 = \frac{[A_3BC]^2}{[A_2]^3\,[B_2][C]^2} \qquad \ldots\ldots\ldots\ldots (1)$$

$$K_2 = \frac{[A_2]^{3/2}\,[B_2]^{1/2}\,[C]}{[A_3BC]}$$

$$\Rightarrow \quad K_2^2 = \frac{[A_2]^3\,[B_2][C]^2}{[A_3BC]^2} \qquad \ldots\ldots\ldots\ldots (2)$$

Comparing (1) & (2)

$$K_2^2 = \frac{1}{K_1}$$

$\Rightarrow K_2 = K_1^{-1/2}$ (option (b))

3. $T_1 = 25 + 273 = 298\ K$

$T_2 = 700\ K$

$$\log\left(\frac{K_2}{K_1}\right) = \frac{\Delta H^o}{2.303\ R}\left(\frac{T_2 - T_1}{T_1 T_2}\right)$$

In this case, $T_2 > T_1$ and $K_1 > K_2$

$$\Rightarrow \frac{2.303\ R \log\left(\frac{K_2}{K_1}\right)}{\left(\frac{T_2 - T_1}{T_1 T_2}\right)} = \Delta H^o$$

$$\Rightarrow \frac{-ve}{+ve} = \Delta H^o$$

$\Delta H^o$ is –ve ie., forward reaction is exothermic (option a)